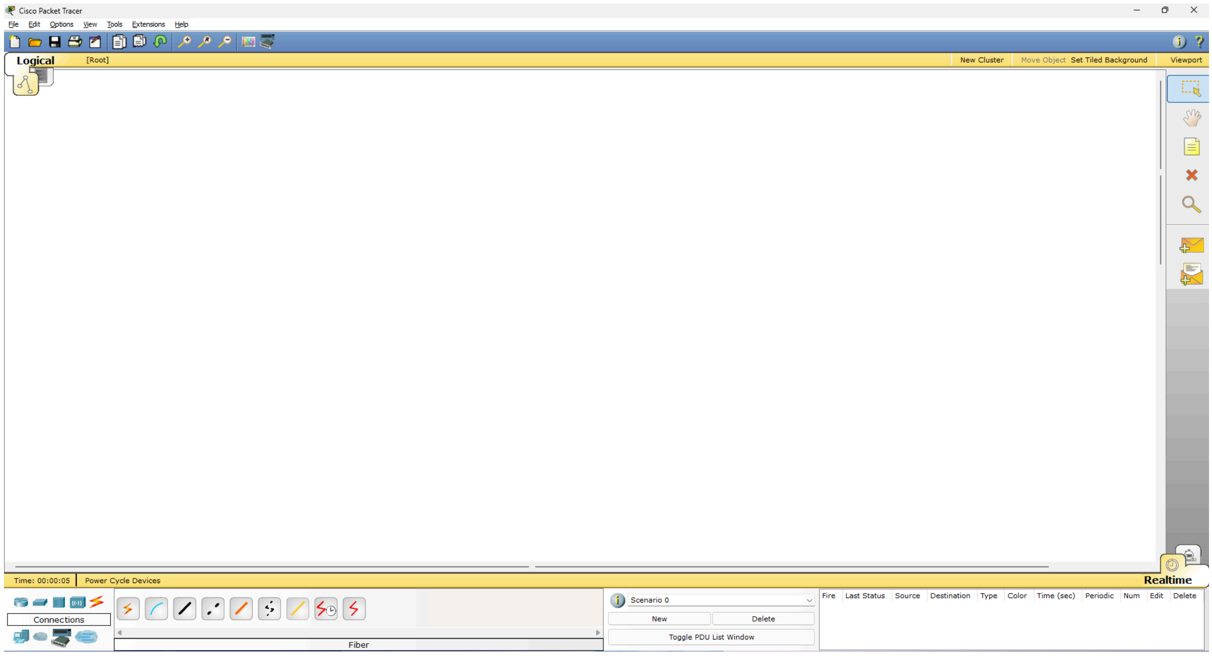
PRACTICALS

Computer Networks Laboratory

<u>Practical 2.</u>

<u>Aim: To implement the topology (atleast two different Network) connected via switch and router in packet tracer simulator.</u>

<u>Step 1:</u> Open Cisco packet tracer 5.1 as shown in **FIG 1**

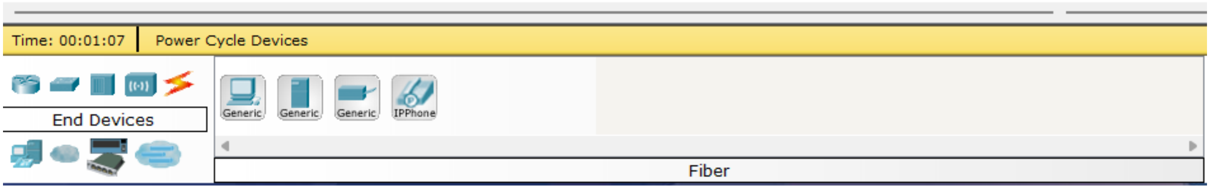


[FIG 1]

<u>Step 2:</u> Select device to make LAN (Local Area Network) Network form the bottom Tool bar as shown in **FIG 2**

There are multiple devices used in this process are follows as:

1. Network Switch
2. Router
3. Hub
4. Wireless device
5. Cables/wires (for connection)
6. End devices (PC, Laptop, Server etc)
7. WAN Enulation
8. Custom made device
9. Multi User connection



[FIG 2]

<u>Step 3:</u> Select PCs, Network Switch, & connecting wires/Cables Shown in **FIG 3**

1. 6 – PCs (3 for each Network)
2. 2 – Network Switch (1 For each Network)
3. 11 – Connecting wires/ cables (5 For Each Network and 1 common For both Network)
4. 2 – Server (1 For Each Network)
5. 2 – Router (1 For Each Network)

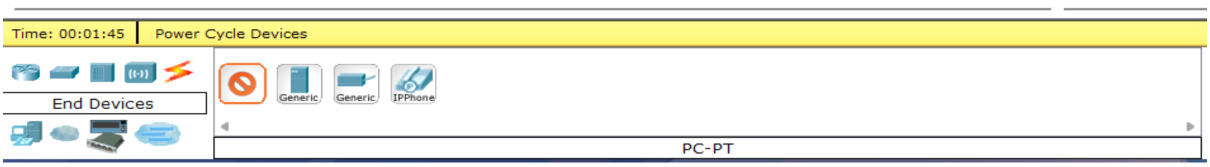


FIG 3

<u>Step 4:</u> Creating Network Using Required devices as shown in **FIG 4**

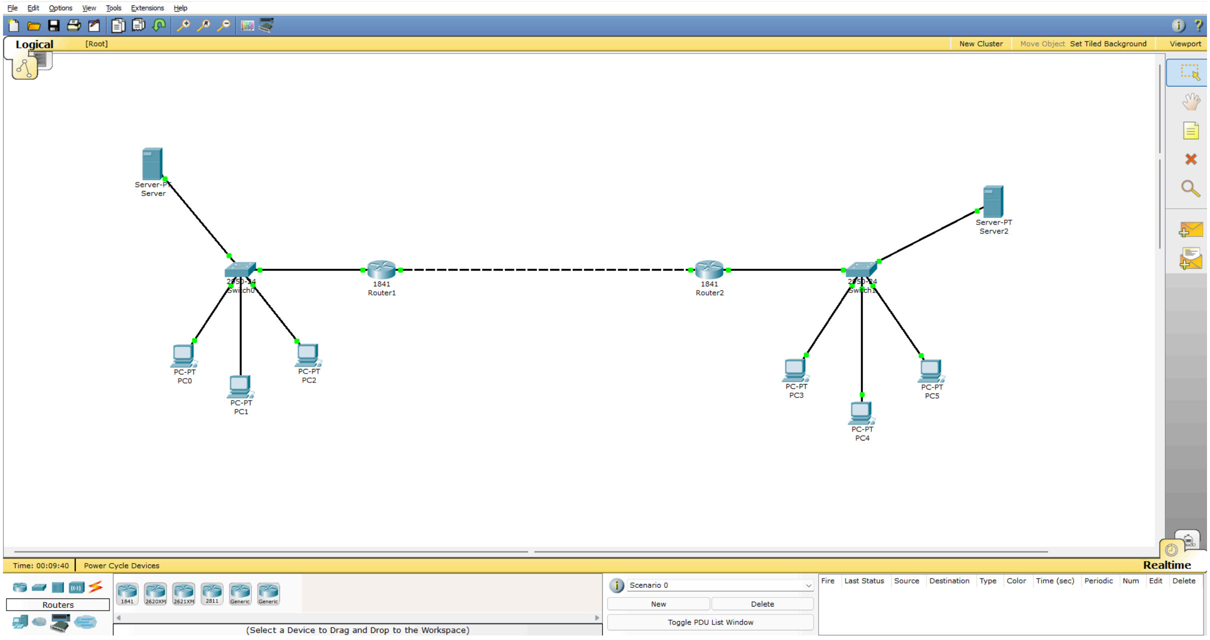


FIG 4

<u>Step 5:</u> Configuring The Setup with Network (DHCP method) Using Default Gateway, Subnet Mask, Ip Address as shown in **FIG 5** and **FIG 6**

<u>Note: This Process Is Only For One Network Do Same Process For All The Networks</u>

1. Click on Server
2. Go to Desktop Tab
3. Click on Ip Configuration

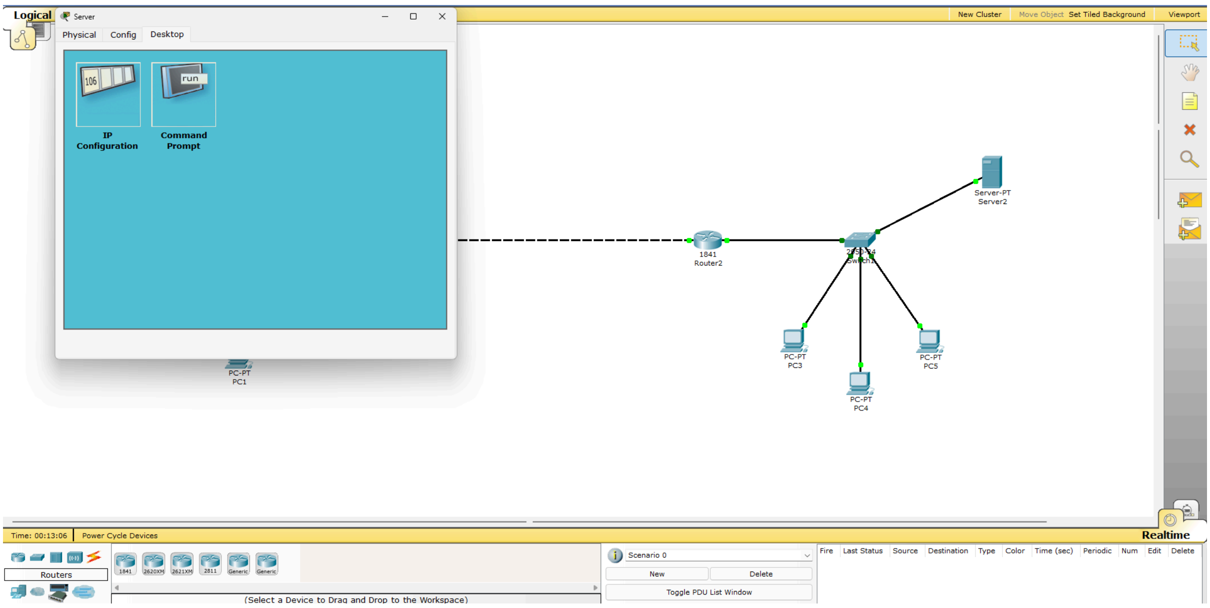


FIG 5

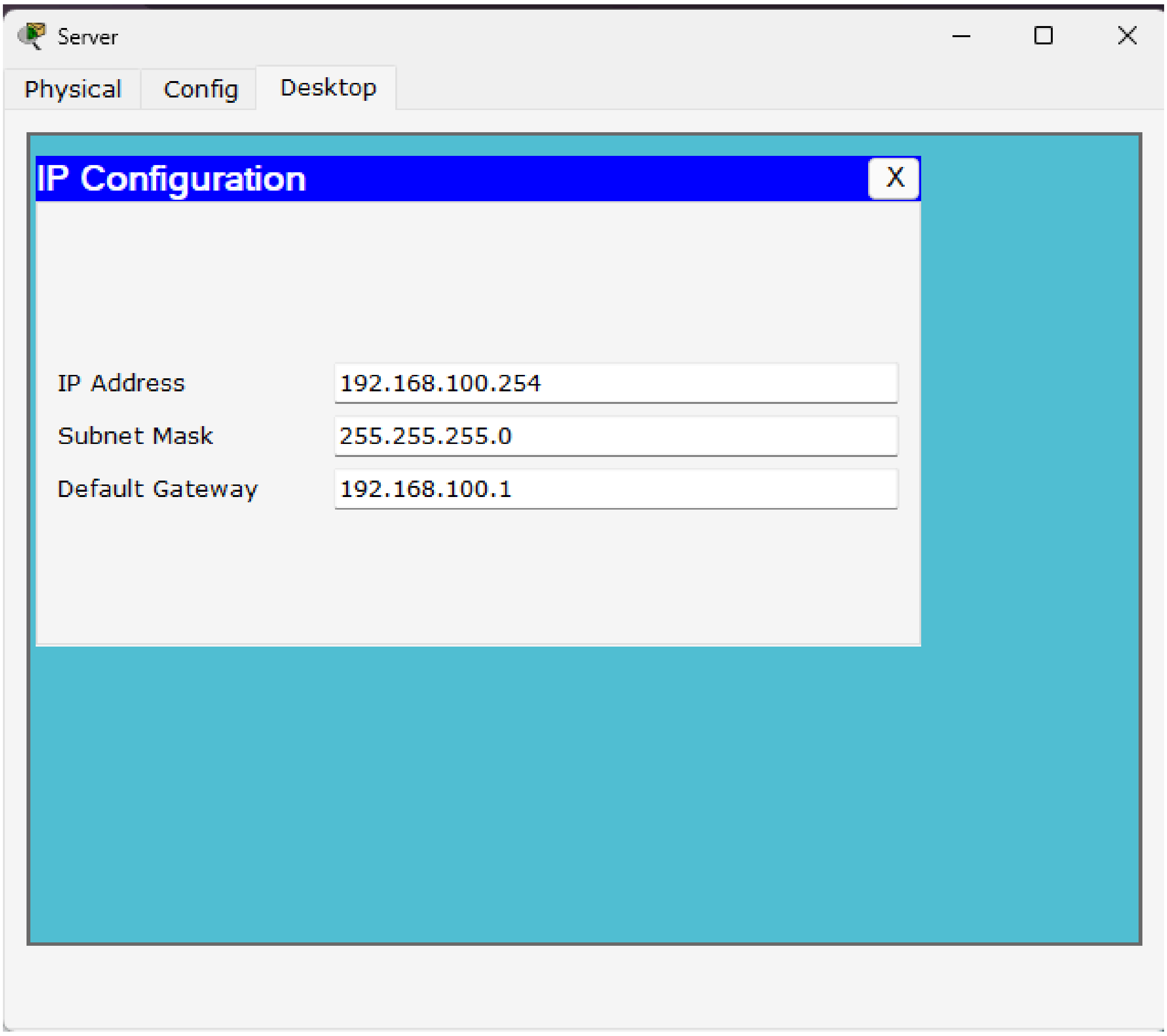


FIG 6

<u>Step 6:</u> Now click on PCs of Same Network Do Following Steps

1. Click on PC
2. Go to Desktop Tab
3. Click on Ip Configuration
4. Select DHCP option
5. All the configuration Details will be fetched directly from Server as Shown in **FIG 7**

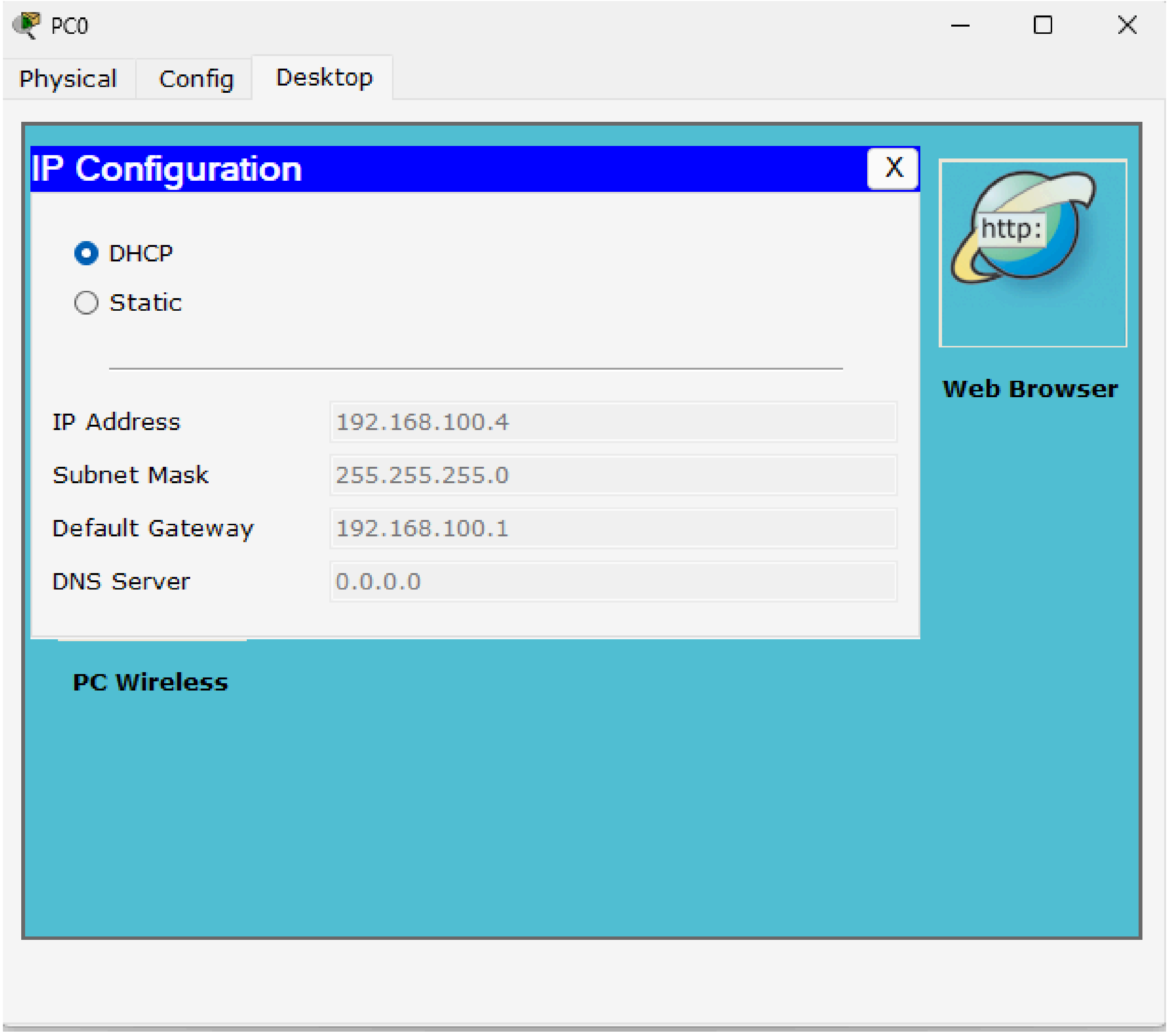

PC0
Physical
Config
Desktop
IP Configuration
X
DHCP
Static
IP Address
192.168.100.4
Subnet Mask
255.255.255.0
Default Gateway
192.168.100.1
DNS Server
0.0.0.0
http:
Web Browser
PC Wireless

FIG 7

Step 7: Now Configuring Routers with the following Commands:
We have to run this commands in CLI of router, To open CLI just click on router and go to CLI and Run the Following Commands

Router1 Configuration (Network 1 side):
Open CLI of Router 1

```
enable
configure terminal

hostname Router1

! LAN interface toward Switch0
interface FastEthernet0/0
 ip address 192.168.100.1 255.255.255.0
 no shutdown
 exit

! WAN link to Router2
interface Serial0/0/0
 ip address 10.0.0.1 255.255.255.252
 clock rate 64000
 no shutdown
 exit

! DHCP pool for Network 1
ip dhcp excluded-address 192.168.100.1
ip dhcp excluded-address 192.168.100.254

ip dhcp pool NETWORK1
 network 192.168.100.0 255.255.255.0
 default-router 192.168.100.1
 dns-server 0.0.0.0
 exit

! Route to reach Network 2
ip route 192.168.200.0 255.255.255.0 10.0.0.2

end
write memory
```

Router2 Configuration (Network 2 side):
Open CLI of Router 2

enable
configure terminal

hostname Router2

! LAN interface toward Switch1
interface FastEthernet0/0
 ip address 192.168.200.1 255.255.255.0
 no shutdown
 exit

! WAN link to Router1
interface Serial0/0/0
 ip address 10.0.0.2 255.255.255.252
 no shutdown
 exit

! DHCP pool for Network 2
ip dhcp excluded-address 192.168.200.1
ip dhcp excluded-address 192.168.200.254

ip dhcp pool NETWORK2
 network 192.168.200.0 255.255.255.0
 default-router 192.168.200.1
 dns-server 0.0.0.0
 exit

! Route to reach Network 1
ip route 192.168.100.0 255.255.255.0 10.0.0.1

end
write memory

<u>Step 7:</u> Try to send Packets Through-out the network

At first attempt it will failed (to share outside the network) but in second attempt of sending packets it will success as shown in **FIG8**

Fire	Last Status	Source	Destination	Type	Color	Time (sec)	Periodic	Num	Edit	Delet
	Failed	Server	PC3	ICMP		0.000	N	0	(edit)	(delet
	Successful	Server	PC3	ICMP		0.000	N	1	(edit)	(delet

FIG 8